Recent Advances in Missing Value Imputation: A Short Survey of State-of-the-Art Techniques

Hyejin Baek

bhj4208@gmail.com

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To Do List from Previous Seminar

23.10.30

- **실험 결과 표 재구성**
  - ✓ 앙상블 방법이 많아 하나의 표로 보이기에는 가독성이 떨어지는 것으로 판단하여 single, 2개 앙상블, 3개 이상 앙상블한 방법으로 분류하여 성능 표 재구성 진행함.
  - ✓ 앙상블 개수로 나누어 표를 보니 가독성이 좋다는 것을 느껴 교수님께 보고 드린 후 피드백 반영 예정임.
  - ✓ table transpose 및 Avg.rank 추가 완료
- **데이터셋 추가**
  - ✓ 7개 추가하여 총 17개 데이터셋 중 6개 데이터셋에서 성능이 우수한 method 발견함.
  - ✓ 실험 결과는 6개 데이터셋에서 우수한 method (dynamic+datawig+zero+knn) 위주로 구성함.
  - ✓ 전체 진행했던 실험 결과(Appendix)에 대해 가독성 있게 표현하지 못해 임시로 표 형태로 업로드 함.

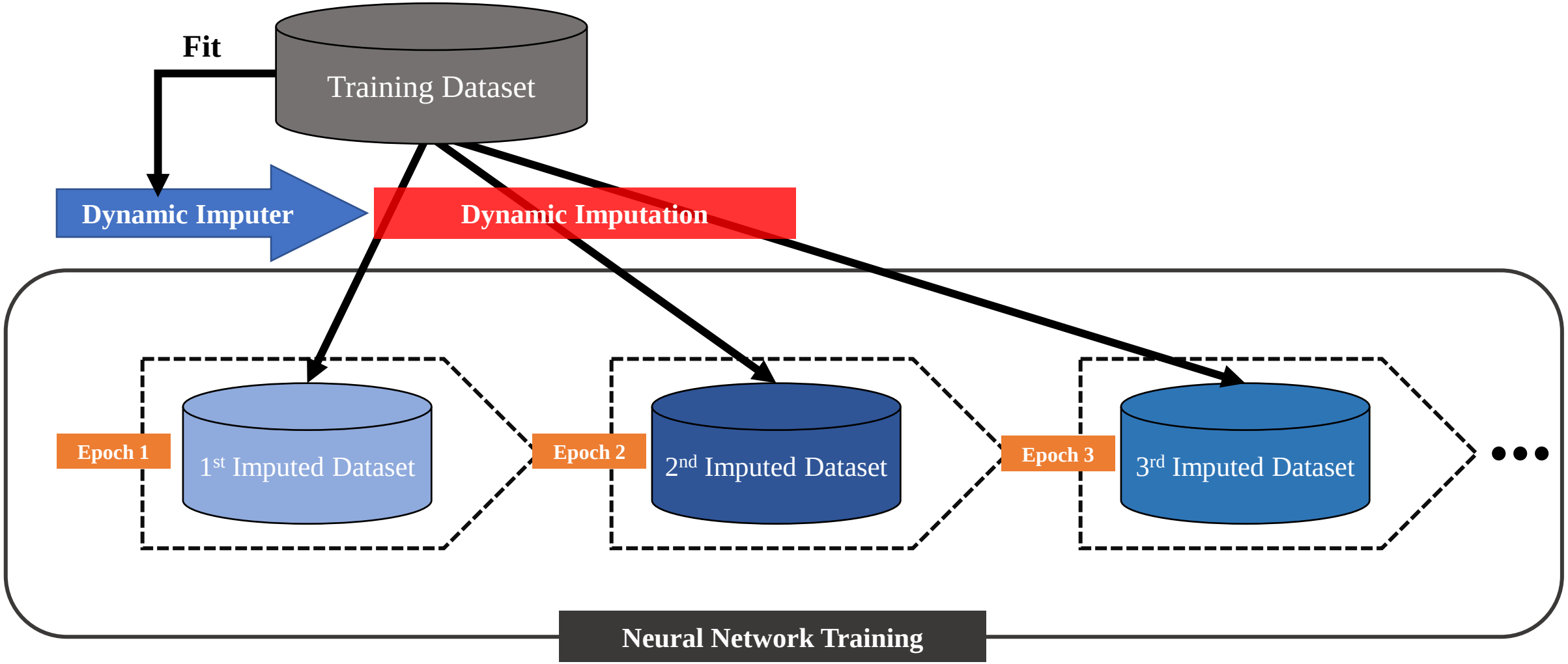
Texonomy

Imputation Method

Machine Learning-based Imputation	Deep Learning-based Imputation	Ensemble-based Imputation
K-means centroid [29]	DataWig [4]	rMisbeta [35]
Feature-wise selection [31]	MLP & DBN [20]	ZIPLN [1]
Takagi-Sugeno (TS) modeling [17]	End-to-end model [44]	Random forest-based + GRU [14]
GRLSR [5]	DAE [32]	Multiple singe imputation [46]
FAIA [10]	VISL [24]	
Clustering & top K nearest neighbor [8]	DeepIn [18]	
Query selection [11]	IFGAN [28]	
MLP [15]	Autoencoder [9, 7]	
Purity-based K nearest neighbors [6], [47], [48]	NAOMI [21]	
	Dynamic[12]	
	CMSDAE [34]	
	GRUF [40]	
	MGAIN [45]	

Dynamic imputation method[1]

✓ Unlike conventional approaches, the dynamic imputation method performs a new missing value imputation with a dynamic imputer for each training epoch.



DataWig[2]

- ✓ The imputation model in DataWig follows the approach of MICE, also referred to as fully conditional specification.
- ✓ For each to-be-imputed column(output column), the user can specify the columns which might contain useful information for imputation(input column)
- ✓ DataWig has a number of features that help to automate end-to-end imputation for practitioners.
- ✓ The data types are detected using heuristics and the corresponding features are learned automatically during the training of the imputation model.
- ✓ All hyperparameters and neural architectures are optimized using random search.

Experimental Settings

- ✓ Dataset : UCI Repository, total 10
- ✓ Dataset description

Table 2: Datasets

Dataset	Domain	Instances	Attributes	Type
adult	Social	32561	15	Categorical
wine	Social	178	13	Integer, Real
iris	Biology	149	5	Real
darwin	Health, Medicine	174	451	Categorical
rice	Biology	3809	8	Real
image-seg	Life	210	20	Real

Experimental Settings

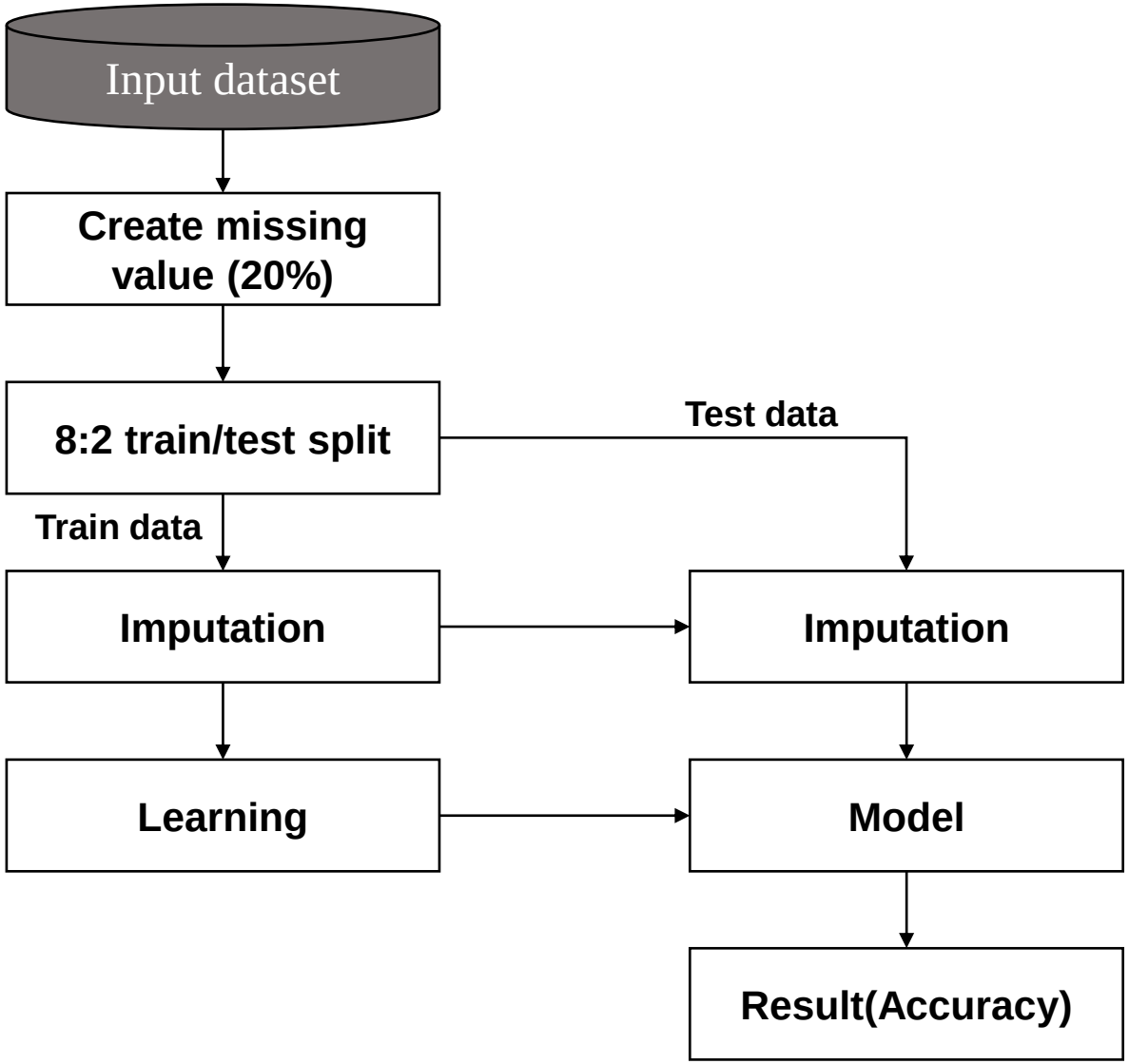
- ✓ Generate 20% missing values randomly in the entire dataset and proceed.
- ✓ Model architecture(2 hidden layer, 1 output layer)[1]

  → Optimizer : Adam

  → loss : BCE

  → Batch size / Epochs : 32/50
- ✓ Measure: Accuracy(8:2 train-test split, 10 iteration)

Experiment Process



Experimental Results

✓ Single Ensemble Imputation method

Table 3: Performance of Single Imputation Method

Dataset	Datawig	Dynamic	Knn	Zero
adult	0.7689 ± 0.0044	0.0850 ± 0.0041	0.7697 ± 0.0046	**0.7727 ± 0.0046** ✓
wine	0.3833 ± 0.0928	**0.4361 ± 0.0938** ✓	0.3833 ± 0.0928	0.3833 ± 0.0928
iris	0.6800 ± 0.0653	0.5500 ± 0.0601	**0.6900 ± 0.0667** ✓	0.6633 ± 0.0674
darwin	0.6829 ± 0.0632	0.4543 ± 0.0803	**0.7086 ± 0.0754** ✓	0.6486 ± 0.1046
rice	0.5667 ± 0.0151	0.5601 ± 0.0117	**0.5671 ± 0.0150** ✓	0.5656 ± 0.0166
image	0.1571 ± 0.0595	**0.1833 ± 0.0621** ✓	0.1571 ± 0.0595	0.1571 ± 0.0595

Experimental Results

✓ Two Ensemble Imputation method

Table 4: Performance of Two Ensemble Imputation Method

Dataset	Zero+Knn	Zero+Dynamic	Zero+Datawig	Knn+Dynamic	Knn+Datawig	Dynamic+Datawig
adult	0.7720 ± 0.0066	0.7873 ± 0.0222	0.7703 ± 0.0039	**0.7889 ± 0.0227** ✓	0.7696 ± 0.0053	0.7862 ± 0.0211
wine	0.3833 ± 0.0928	0.7931 ± 0.1665	0.3833 ± 0.0928	**0.8000 ± 0.1729** ✓	0.3833 ± 0.0928	0.8000 ± 0.1727
iris	0.6467 ± 0.0872	0.7567 ± 0.1131	0.6533 ± 0.0694	**0.7700 ± 0.1264** ✓	0.6833 ± 0.0679	0.7400 ± 0.1083
darwin	0.6329 ± 0.0608	0.6614 ± 0.2175	0.6614 ± 0.0944	0.6414 ± 0.1987	0.6614 ± 0.0752	**0.6514 ± 0.2087** ✓
rice	0.5659 ± 0.0149	0.7394 ± 0.1794	0.5665 ± 0.0149	**0.7425 ± 0.1822** ✓	0.5675 ± 0.0155	0.7402 ± 0.1802
image	0.1571 ± 0.0595	0.8560 ± 0.0535	0.1571 ± 0.0595	**0.8619 ± 0.0503** ✓	0.1571 ± 0.0595	0.8571 ± 0.0578

Experimental Results

✓ Three or Four Ensemble Imputation method

Table 5: Performance of Three or Four Ensemble Imputation Method

Dataset	dynamic+datawig+zero	dynamic+datawig+knn	datawig+zero+knn	dynamic+zero+knn	all
adult	0.7949 ± 0.0212	0.7965 ± 0.0223	0.7700 ± 0.0048	0.7963 ± 0.0214	**0.7983 ± 0.0200** ✓
wine	0.8491 ± 0.1584	0.8500 ± 0.1589	0.3833 ± 0.0928	0.8500 ± 0.1597	**0.8750 ± 0.1457** ✓
iris	0.7722 ± 0.1004	0.7856 ± 0.1121	0.6633 ± 0.0727	0.7811 ± 0.1081	**0.7858 ± 0.0960** ✓
darwin	0.7171 ± 0.1969	0.7152 ± 0.1952	0.6676 ± 0.0852	0.7162 ± 0.1950	**0.7479 ± 0.1802** ✓
rice	0.7987 ± 0.1691	0.8008 ± 0.1703	0.5603 ± 0.0301	0.8010 ± 0.1705	**0.8304 ± 0.1564** ✓
image	0.8571 ± 0.0514	0.8643 ± 0.0492	0.1571 ± 0.0595	0.8579 ± 0.0490	**0.8673 ± 0.0435** ✓

Experimental Results

- ✓ 앙상블 실험 진행 결과, 2개의 method를 앙상블한 결과에서는 knn+dynamic의 성능이 높음
- ✓ 3개 이상의 method를 앙상블한 결과에서는 datawig+dynamic+knn+zero로써 4가지 method를 모두 사용한 method가 성능이 높게 나옴.
- ✓ 모든 method를 비교했을 때 4가지 method를 앙상블한 결과가 가장 높은 성능이 나옴을 확인할 수 있음.

Appendix

✓ 전체 실험 결과(양이 너무 많아 임시로 엑셀 파일로 보고 드립니다)

	Datawig	Dynamic	Knn	Zero	zero+knn	zero+dynamic	zero+datawig	knn+dynamic	knn+datawig	dynamic+datawig	dynamic+datawig+zero	dynamic+datawig+knn	datawig+zero+knn	dynamic+zero+knn	all
breast	0.6965 ± 0.0671	0.7053 ± 0.0542	0.7193 ± 0.0360	0.6649 ± 0.0574	0.6860 ± 0.0732	0.6904 ± 0.0608	0.7018 ± 0.0444	0.6868 ± 0.0488	**0.7351 ± 0.0350**	0.7017 ± 0.0401	0.7040 ± 0.0531	0.6989 ± 0.0471	0.7082 ± 0.0465	0.6836 ± 0.0600	0.7052 ± 0.0444
heart	0.3869 ± 0.1164	0.5492 ± 0.0738	0.3705 ± 0.0752	0.3770 ± 0.0758	0.4189 ± 0.0840	0.7877 ± 0.0621	0.4098 ± 0.0887	**0.7943 ± 0.0555**	0.4057 ± 0.0977	0.7918 ± 0.0596	0.7842 ± 0.0597	0.7803 ± 0.0607	0.4235 ± 0.0690	0.7814 ± 0.0587	0.7717 ± 0.0608
adult	0.7689 ± 0.0044	0.0850 ± 0.0041	0.7697 ± 0.0046	0.7727 ± 0.0046	0.7720 ± 0.0066	0.7873 ± 0.0222	0.7703 ± 0.0039	0.7889 ± 0.0227	0.7696 ± 0.0053	0.7862 ± 0.0211	0.7949 ± 0.0212	0.7965 ± 0.0223	0.7700 ± 0.0048	0.7963 ± 0.0214	**0.7943 ± 0.0555**
wine	0.3833 ± 0.0928	0.4361 ± 0.0938	0.3833 ± 0.0928	0.3833 ± 0.0928	0.3833 ± 0.0928	0.7931 ± 0.1665	0.3833 ± 0.0928	0.8000 ± 0.1729	0.3833 ± 0.0928	0.8000 ± 0.1727	0.8491 ± 0.1584	0.8500 ± 0.1589	0.3833 ± 0.0928	0.8500 ± 0.1597	**0.7943 ± 0.0555**
spambase	0.9060 ± 0.0099	0.6032 ± 0.0129	**0.7943 ± 0.0555**	0.9162 ± 0.0063	0.9187 ± 0.0196	0.7566 ± 0.1538	0.9025 ± 0.0169	0.7542 ± 0.1514	0.9176 ± 0.0168	0.7548 ± 0.1521	0.8085 ± 0.1455	0.8059 ± 0.1437	0.3833 ± 0.0928	0.8059 ± 0.1437	0.8307 ± 0.1318
chess	0.0503 ± 0.0030	0.2109 ± 0.0059	0.0503 ± 0.0030	0.0503 ± 0.0030	0.0503 ± 0.0030	0.9508 ± 0.0028	0.0503 ± 0.0030	0.9502 ± 0.0032	0.0503 ± 0.0030	**0.7943 ± 0.0555**	0.9507 ± 0.0028	0.9500 ± 0.0030	0.0503 ± 0.0030	0.0503 ± 0.0030	0.9507 ± 0.0035
post patient	0.0944 ± 0.0500	0.6111 ± 0.0930	0.1889 ± 0.0903	0.1500 ± 0.0705	0.1778 ± 0.1160	0.9639 ± 0.0474	0.1389 ± 0.0795	0.9500 ± 0.0524	0.1278 ± 0.0964	0.9611 ± 0.0500	0.9667 ± 0.0421	0.9648 ± 0.0442	0.1185 ± 0.0857	**0.9722 ± 0.0448**	0.9694 ± 0.0447
liver	0.0638 ± 0.0312	0.1145 ± 0.0412	0.0638 ± 0.0312	0.0638 ± 0.0312	0.0638 ± 0.0312	0.9348 ± 0.0319	0.0638 ± 0.0312	**0.9362 ± 0.0312**	0.0638 ± 0.0312	0.9348 ± 0.0302	0.9348 ± 0.0321	0.9338 ± 0.0310	0.0638 ± 0.0312	0.9343 ± 0.0312	0.9355 ± 0.0311
abalone	0.3157 ± 0.0187	0.4867 ± 0.0090	0.3121 ± 0.0126	0.3087 ± 0.0120	0.3134 ± 0.0149	**0.6898 ± 0.0114**	0.3115 ± 0.0151	0.6884 ± 0.0146	0.3106 ± 0.0145	0.6897 ± 0.0126	0.6850 ± 0.0137	0.6841 ± 0.0146	0.3138 ± 0.0145	0.6854 ± 0.0157	0.6851 ± 0.0160
lymphography	0.0100 ± 0.0213	0.3867 ± 0.0562	0.0100 ± 0.0213	0.0100 ± 0.0213	0.0100 ± 0.0213	**0.9883 ± 0.0218**	0.0100 ± 0.0213	**0.9883 ± 0.0218**	0.0100 ± 0.0213	0.9867 ± 0.0245	0.9867 ± 0.0221	0.9878 ± 0.0219	0.0100 ± 0.0213	0.9900 ± 0.0213	0.9875 ± 0.0232
Iris	0.6800 ± 0.0653	0.5500 ± 0.0601	0.6900 ± 0.0667	0.6633 ± 0.0674	0.6467 ± 0.0872	0.7567 ± 0.1131	0.6533 ± 0.0694	0.7700 ± 0.1264	0.6833 ± 0.0679	0.7400 ± 0.1083	0.7722 ± 0.1004	0.7856 ± 0.1121	0.6633 ± 0.0727	0.7811 ± 0.1081	**0.7858 ± 0.0960**
darwin	0.6829 ± 0.0632	0.4543 ± 0.0803	0.7086 ± 0.0754	0.6486 ± 0.1046	0.6329 ± 0.0608	0.6614 ± 0.2175	0.6614 ± 0.0944	0.6414 ± 0.1987	0.6614 ± 0.0752	0.6514 ± 0.2087	0.7171 ± 0.1969	0.7152 ± 0.1952	0.6676 ± 0.0852	0.7162 ± 0.1950	**0.7479 ± 0.1802**
image seg	0.1571 ± 0.0595	0.1833 ± 0.0621	0.1571 ± 0.0595	0.1571 ± 0.059	0.1571 ± 0.0595	0.8560 ± 0.0535	0.1571 ± 0.0595	0.8619 ± 0.0503	0.1571 ± 0.0595	0.8571 ± 0.0578	0.8571 ± 0.0514	0.8643 ± 0.0492	0.1571 ± 0.0595	0.8579 ± 0.0490	**0.8673 ± 0.0435**
national_health	0.9276 ± 0.0134	0.8636 ± 0.0158	**0.9393 ± 0.0070**	0.8987 ± 0.0122	0.9185 ± 0.0237	0.9137 ± 0.0518	0.9168 ± 0.0185	0.9109 ± 0.0485	0.9310 ± 0.0143	0.9124 ± 0.0498	0.9300 ± 0.0476	0.9280 ± 0.0469	0.9224 ± 0.0197	0.9294 ± 0.0477	0.9353 ± 0.0427
magic	0.7885 ± 0.0067	0.6687 ± 0.0076	0.7985 ± 0.0064	0.7756 ± 0.0085	0.7946 ± 0.0155	0.7405 ± 0.0725	0.7850 ± 0.0094	0.7424 ± 0.0745	**0.7992 ± 0.0102**	0.7413 ± 0.0738	0.7636 ± 0.0685	0.7631 ± 0.0676	0.7919 ± 0.0137	0.7656 ± 0.0693	0.7768 ± 0.0636
rice	0.5667 ± 0.0151	0.5601 ± 0.0117	0.5671 ± 0.0150	0.5656 ± 0.0166	0.5659 ± 0.0149	0.7394 ± 0.1794	0.5665 ± 0.0149	0.7425 ± 0.1822	0.5675 ± 0.0155	0.7402 ± 0.1802	0.7987 ± 0.1691	0.8008 ± 0.1703	0.5603 ± 0.0301	0.8010 ± 0.1705	**0.8304 ± 0.1564**
bank-marketing	0.8789 ± 0.0096	**0.8815 ± 0.0097**	0.8776 ± 0.0130	0.8783 ± 0.0087	0.8772 ± 0.0092	0.8694 ± 0.0173	0.8759 ± 0.0104	0.8690 ± 0.0158	0.8785 ± 0.0101	0.8721 ± 0.0141	0.8691 ± 0.0141	0.8671 ± 0.0161	0.8751 ± 0.0117	0.8623 ± 0.0174	0.8673 ± 0.0143
avg.rank	9.7	10.1	8.2	11.1	10.1	6.8	10.2	6.9	8.7	6.9	5.2	5.8	10.2	5.7	**3.7**

To Do List

- ✓ 실험 결과에 대한 교수님과 논의 필요

Reference

[1] Han, Jongmin, and Seokho Kang, “Dynamic imputation for improved training of neural network with missing values” Expert Systems with Applications, 2022

[2] Biessmann, and Felix, “DataWig:Missing Value Imputation for tables” Journal of Machine Learning, 2019

Thank you